INFLUENCE BOUNDS FOR ARBITRARY PRODUCT MEASURES

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Abstract. Every bounded function of low interaction degree on a finite product space with an arbitrary product measure has a coordinate whose influence is at least a fixed power of its variance, with constants independent of all atom probabilities. As a consequence, low-degree functions under any product measure are approximated by shallow deterministic decision trees. All proofs are complete and self-contained.

1. Introduction

Aaronson and Ambainis showed that a lower bound on the largest coordinate influence of a bounded low-degree polynomial in terms of its variance would allow quantum algorithms with few queries to be simulated classically on most inputs [1]. Most results of this kind are proved for the uniform measure on the Boolean cube, or for product measures whose biases stay away from 0 and 1, and their constants degrade as the smallest atom probability tends to zero. This paper proves such an influence bound for an arbitrary product measure on an arbitrary finite product, with constants that do not depend on the measure at all.

Let $\Omega = \prod_{i=1}^{n} \Omega_i$ be a finite product and let $\mu = \bigotimes_i \mu_i$ be any product probability measure on Ω. Coordinates with zero atom probabilities and coordinates supported on a single point are permitted. The function f is defined on the whole product. For independent $X, Y \sim \mu$, write $X^{[i]} = (X_1, \dots, X_{i-1}, Y_i, X_{i+1}, \dots, X_n)$ and set

$$\mathcal{I}_i(f) = \mathbb{E}|f(X) - f(X^{[i]})|^2, \qquad \mathcal{L}_i(f) = \mathbb{E}|f(X) - f(X^{[i]})|. \tag{1.1}$$

These are independent-resampling quantities. In particular, for fair Boolean inputs, $\mathcal{I}_i$ is one half of the squared bit-flip increment.

Definition 1.1. The interaction degree of f is at most d if there is an expansion

$$f(x) = \sum_{S \subseteq [n],\ |S| \le d} g_S(x_S)$$

with real functions g_S depending on the coordinates in S. For Boolean alphabets this is the usual degree of the unique multilinear polynomial representing f.

Put

$$c_* = \frac{16\sqrt{15}}{125\pi^4}, \qquad b = \max\left(1, \left\lceil \log_2 \max_i |\Omega_i| \right\rceil\right). \tag{1.2}$$

Theorem 1.2. *Let $f : \Omega \to [0,1]$ have interaction degree at most $d \ge 1$. If $v = \operatorname{Var}_\mu f > 0$, then some coordinate satisfies*

$$\mathcal{I}_i(f) \ge c_* \frac{v^{5/2}}{d^6 b^4}. \tag{1.3}$$

The constant is independent of n, of the measures μ_i, and of their smallest positive atom probabilities.

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Corollary 1.3. *Under the hypotheses of Theorem 1.2, for every $0 < \varepsilon < 1/2$ there exists a deterministic decision tree of depth at most*

$$\min\left(n, \left\lceil \frac{256}{c_*} d^7 b^4 \varepsilon^{-3} \right\rceil\right)$$

whose real output $A(X)$ satisfies $\mathbb{E}_\mu (A(X) - f(X))^2 \le \varepsilon^2$. Each query reveals one complete coordinate in Ω_i.

Two structural results organize the argument. We state them here and give complete proofs in Appendices A and B, including the analytic estimate needed for the certificate bound. For a Boolean cube $Q_k = \{0,1\}^k$ and a function h, a row at u is a nonnegative vector a^u such that

$$|h(u) - h(z)| \le \sum_{j : u_j \ne z_j} a^u_j \quad (z \in Q_k).$$

For Boolean h, this says that every block changing its value has weight at least one.

Proposition 1.4 (Turturean's antipodal inequality)**.** *Let $h\colon Q_k \to \{0,1\}$ have rows a^u. Let $\bar{u}$ denote the bitwise complement and let $B_j(h)$ count the unoriented edges in direction j whose endpoints have different values. For all $w_j > 0$,*

$$\sum_{u : h(u)=1,\ h(\bar{u})=0} \frac{1}{\sum_j w_j a^u_j a^{\bar{u}}_j} \le \sum_j \frac{B_j(h)}{w_j}. \tag{1.4}$$

Every denominator on the left is positive.

Proposition 1.4 and its determinant method are due to Turturean [6]. Appendix B gives a complete proof from a determinant of agreement monomials, so the present argument does not depend on any unpublished source.[1]

Proposition 1.5 (Agarwal–Ben-David's row bound)**.** *If $F\colon Q_N \to [0,1]$ has polynomial degree at most D, it has a row at every input, each of cost at most $\pi^2 D^2/4$.*

This is the row estimate of Agarwal and Ben-David [2]. Appendix A proves it using trigonometric interpolation, Chebyshev polynomials and a finite-dimensional separation argument. The product transfer, alphabet reduction and query estimate are proved in the main text. No step assumes an unproved conjecture.

Aaronson and Ambainis established the connection between polynomial influences and quantum query simulation [1]. Girish, Mittal, Nehoran and Raz formulate distributional variants and prove equivalences for fixed biases [5]. Their Remark 3.5 also treats coordinate-dependent biases in a fixed interval $[a, 1-a]$. Our statement includes arbitrary choices of the probabilities at each input length, with no such parameter in the bound.

2. Conditioning two inputs on unordered pairs

For $x, y \in \Omega$ put $\Delta(x,y) = \{i : x_i \ne y_i\}$. A row for f at $x \in \Omega$ will mean a vector $a^x \ge 0$ with

$$|f(x) - f(y)| \le \sum_{i \in \Delta(x,y)} a^x_i \quad (y \in \Omega). \tag{2.1}$$

Lemma 2.1 (Product antipodal transfer)**.** *Let $g\colon \Omega \to \{0,1\}$ have rows a^x. For any positive weights $w = (w_i)$,*

$$\mathbb{E}_{X,Y} \frac{\mathbf{1}_{\{g(X)=1, g(Y)=0\}}}{\sum_{i \in \Delta(X,Y)} w_i a^X_i a^Y_i} \le \frac{1}{2} \sum_i \frac{\mathcal{L}_i(g)}{w_i}. \tag{2.2}$$

The integrand is defined to be zero when its numerator is zero.

[1] The proof is given in full so that every step used here can be checked within this paper.

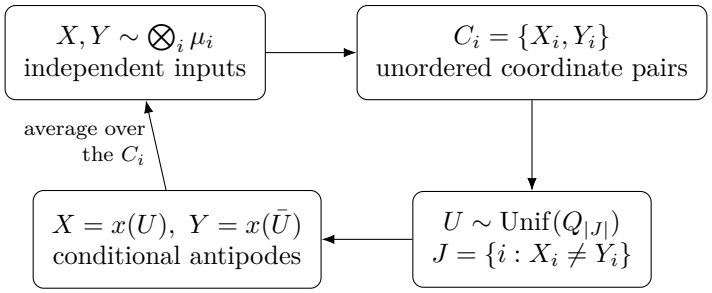


FIGURE 1. The orientations are fair even when the original atom probabilities are arbitrarily small. Averaging the weighted inequality precedes the choice of an influential coordinate.

Proof. Condition on the unordered pair $C_i = \{X_i, Y_i\}$ at every coordinate, retaining multiplicity when its entries agree. The C_i are independent. For $C_i = \{s, t\}$ with $s \neq t$, the two orientations have equal conditional probabilities because $\mu_i(s)\mu_i(t) = \mu_i(t)\mu_i(s)$. Consequently, after conditioning, the unequal coordinates form a fair Boolean cube. Its uniformly chosen vertex represents X and its antipode represents Y. The other coordinates are fixed.

Restrict the rows to the unequal coordinates. They remain valid because every pair in this cube is also a pair in Ω. Apply (1.4) and divide by the cube size. The normalized number of boundary edges in a direction is one half of the probability that flipping that direction changes g.

Now average over the C_i. The left side becomes the left side of (2.2). Flipping one conditional orientation replaces X_i by Y_i, leaving X_{-i} fixed. Before conditioning, Y_i is an independent sample from μ_i, so the averaged boundary probability is $\mathcal{L}_i(g)$. If the conditional cube is a singleton both sides of its inequality vanish. This proves the claim, including degenerate measures. □

Lemma 2.2 (Real-valued pair inequality)**.** *Let $f\colon \Omega \to \mathbb{R}$ have rows a^x, and put $s_{xy}(w) = \sum_{i \in \Delta(x,y)} w_i a_i^x a_i^y$. Whenever $f(x) > f(y)$ this denominator is positive. Moreover,*

$$\mathbb{E}_{X,Y}\left[\mathbf{1}_{\{f(X)>f(Y)\}} \frac{(f(X)-f(Y))^3}{6 s_{XY}(w)}\right] \leq \frac{1}{2}\sum_i \frac{\mathcal{L}_i(f)}{w_i}. \tag{2.3}$$

Proof. For a real number t not equal to a value of f, let $g_t(x) = \mathbf{1}_{\{f(x)>t\}}$ and divide the row a^x by $|f(x) - t|$. If $g_t(x) \neq g_t(y)$, then $|f(x) - f(y)| \geq |f(x) - t|$, so the divided vector is a row for g_t. Apply Lemma 2.1. For $f(x) > f(y)$ its summand is

$$\mathbf{1}_{\{f(y)<t<f(x)\}} \frac{(f(x)-t)(t-f(y))}{s_{xy}(w)}.$$

The integral of its numerator over t equals $(f(x) - f(y))^3/6$. Also $\int \mathcal{L}_i(g_t)\, dt = \mathcal{L}_i(f)$. Integrating the finite sums gives (2.3).

For positivity, the divided rows of a Boolean function at differently valued inputs have positive overlap on some changed coordinate. Indeed, partition the changed coordinates according to which of the two row weights is smaller. The intermediate input using one part forces one of the two sums of smaller weights to be at least one. This also proves positivity of $s_{xy}(w)$. □

3. Removing the weights and selecting one coordinate

Lemma 3.1 (A finite-measure calculation)**.** *Let A, B carry positive finite weights α_x, β_y, and let λ^x, ν^y be nonnegative vectors. Suppose the quantities $s_{xy} = \sum_{i \in J(x,y)} w_i \lambda_i^x \nu_i^y$ are positive, and for every $w_i > 0$,*

$$\sum_{x,y} \frac{\alpha_x \beta_y}{s_{xy}} \leq C \sum_i \frac{1}{w_i}.$$

Then

$$\left(\sum_x \frac{\alpha_x}{\|\lambda^x\|_1}\right)\left(\sum_y \frac{\beta_y}{\|\nu^y\|_1}\right) \le C. \tag{3.1}$$

Proof. The row norms are positive by the assumption on s_{xy}. Write $v_x = 1/\|\lambda^x\|_1$, $u_y = 1/\|\nu^y\|_1$, $A_0 = \sum_x \alpha_x v_x$, $B_0 = \sum_y \beta_y u_y$, and

$$P_i = \sum_x \alpha_x v_x^2 \lambda_i^x, \qquad Q_i = \sum_y \beta_y u_y^2 \nu_i^y.$$

Cauchy–Schwarz over the pairs gives

$$(A_0 B_0)^2 \le C\left(\sum_i w_i P_i Q_i\right)\left(\sum_i w_i^{-1}\right).$$

Here we enlarged $J(x,y)$ to all coordinates in the first factor, whose terms are nonnegative. Choose $w_i = (\sqrt{P_i Q_i} + \eta)^{-1}$ and let $\eta \downarrow 0$. The right side tends to $C(\sum_i \sqrt{P_i Q_i})^2 \le C(\sum_i P_i)(\sum_i Q_i) = CA_0B_0$. Division by $A_0 B_0 > 0$ proves the lemma. □

Proposition 3.2 (A product certificate bound)**.** *Suppose $f\colon \Omega \to \mathbb{R}$ has variance $v > 0$ and rows of cost at most K at every point. Put $L = \sum_i \mathcal{L}_i(f)$. Then $0 < L \le K$, and*

$$\max_i \mathcal{I}_i(f) \ge \frac{\sqrt{15}}{125}\,\frac{v^{5/2}}{LK^2}. \tag{3.2}$$

Proof. For $M = \max_i \mathcal{L}_i(f)$ and any $r \in \mathbb{R}$, restrict (2.3) to $A = \{f > r\}$ and $B = \{f \le r\}$. Replacing $(f(x) - f(y))^3$ by $(f(x) - r)^3$ and multiplying by six gives

$$\sum_{x\in A, y\in B} \mu(x)\mu(y)\frac{(f(x)-r)^3}{s_{xy}(w)} \le 3M \sum_i w_i^{-1}.$$

Use Lemma 3.1 with $\lambda^x = a^x/(f(x)-r)^3$ and $\nu^y = a^y$. The norms are at most $K/(f(x)-r)^3$ and K. Hence

$$\mathbb{E}(f-r)_+^3 \Pr(f \le r) \le 3K^2 M. \tag{3.3}$$

Empty sets cause no difficulty. Applying the same statement to $-f$ and choosing a median m gives $\mathbb{E}|f-m|^3 \le 12K^2M$. Since $v \le \mathbb{E}(f-m)^2 \le (\mathbb{E}|f-m|^3)^{2/3}$, we obtain

$$\max_i \mathcal{L}_i(f) \ge \frac{v^{3/2}}{12K^2}. \tag{3.4}$$

For each fixed x, averaging the single-coordinate row constraints over independent replacements shows $\sum_i \mathbb{E}_{Y_i}|f(x) - f(x_{-i}, Y_i)| \le \sum_i a_i^x \le K$. Thus $L \le K$. If $L = 0$, successive averaging over all coordinates makes f constant almost surely, contrary to $v > 0$.

Let

$$R = \{i : 5L\mathcal{I}_i(f) < 4v\mathcal{L}_i(f)\}, \qquad h = \mathbb{E}[f \mid X_{[n]\setminus R}].$$

Then $\sum_{i\in R} \mathcal{I}_i(f) \le 4v/5$. Averaging one coordinate loses variance $\mathcal{I}_i/2$, and Jensen's inequality shows that it increases neither resampling quantity of any remaining coordinate. It follows that $\operatorname{Var} h \ge 3v/5$.

Averaging also preserves the row bound. To see this, average the row at (x_{-R}, z_R) over z_R, retain its entries outside R, and set its entries in R to zero. Using the same z_R in a comparison of two inputs proves (2.1), with cost at most K. Apply (3.4) to h. It produces a coordinate $j \notin R$ such that

$$\mathcal{L}_j(f) \ge \mathcal{L}_j(h) \ge \frac{(3v/5)^{3/2}}{12K^2}.$$

The definition of R therefore gives

$$\mathcal{I}_j(f) \geq \frac{4v}{5L}\mathcal{L}_j(f) \geq \frac{4}{60}\left(\frac{3}{5}\right)^{3/2}\frac{v^{5/2}}{LK^2} = \frac{\sqrt{15}}{125}\frac{v^{5/2}}{LK^2}.$$

□

The median and averaging steps follow Turturean's strategy. The product pair transfer is what makes their constants independent of the measures.

4. Degree estimates and the alphabet reduction

Lemma 4.1. *If $G\colon Q_k \to [0,1]$ has degree at most d, then at every vertex z,*

$$\sum_i |G(z) - G(z^{\oplus i})| \leq d^2. \tag{4.1}$$

Proof. Extend $P(s) = 2G((s+\mathbf{1})/2) - 1$ multilinearly to $[-1,1]^k$. It takes values in $[-1,1]$ there. At a vertex $s = 2z - \mathbf{1}$,

$$\sum_i |G(z) - G(z^{\oplus i})| = \sum_i |\partial_i P(s)|.$$

Choose y_i to be the sign of $\partial_i P(s)$. Lemma A.2 bounds $|DP(s)[y]|$ by d^2, which is the desired sum. This is the pointwise inequality of Filmus, Hatami, Keller and Lifshitz [4], with the normalization used here. □

Lemma 4.2. *For every $f\colon \Omega \to [0,1]$ of interaction degree at most d,*

$$\sum_i \mathcal{L}_i(f) \leq d^2.$$

Proof. For fixed endpoints x, y, restrict f to the cube which chooses either x_i or y_i at each coordinate. Its Boolean polynomial degree is at most d, because every component g_S depends on at most d of the choice bits. Apply Lemma 4.1 at the vertex corresponding to x. Averaging over independent $x, y \sim \mu$ gives the claim exactly. □

Lemma 4.3. *If $f\colon \Omega \to [0,1]$ has interaction degree at most d, then it has rows on Ω of cost at most $\pi^2 d^2 b^2/4$.*

Proof. Choose a surjection $D_i\colon \{0,1\}^b \to \Omega_i$ at each coordinate, and let $D = (D_1, \dots, D_n)$. The Boolean function $F = f \circ D$ takes values in $[0,1]$ everywhere. Each component $g_S \circ D$ depends on at most $b|S|$ Boolean bits, so its degree is at most bd. Thus Proposition 1.5 supplies rows α^u for F of cost at most $\pi^2 d^2 b^2/4$.

Fix a section $e_i\colon \Omega_i \to \{0,1\}^b$ of D_i. At $x \in \Omega$ define $a_i^x = \sum_{j=1}^b \alpha_{ij}^{e(x)}$. For any y, the codes $e(x), e(y)$ agree outside the blocks $i \in \Delta(x,y)$. The Boolean row inequality bounds $|F(e(x)) - F(e(y))| = |f(x) - f(y)|$ by the sum of these block weights. The total cost equals the cost of $\alpha^{e(x)}$, proving the lemma. The codes are used only for this certificate argument. They impose no condition on μ. □

Proof of Theorem 1.2. Insert $L \leq d^2$ and $K \leq \pi^2 d^2 b^2/4$ into (3.2). Their product satisfies $LK^2 \leq \pi^4 d^6 b^4/16$, giving (1.3). □

5. Decision trees and quantum queries

We record the product orthogonal decomposition explicitly. For every coordinate, take an orthonormal basis of $L^2(\mu_i)$ whose first element is the constant one. Products of these basis elements form an orthonormal basis of $L^2(\mu)$. Grouping by the set of coordinates with nonconstant basis elements gives

$$f = \sum_{S \subseteq [n]} f_S, \qquad \mathbb{E}[f_S \mid X_{[n]\setminus\{i\}}] = 0 \quad (i \in S).$$

All summands with $|S| > d$ vanish for a function of interaction degree d. Parseval and independent resampling yield

$$\operatorname{Var} f = \sum_{S \neq \varnothing} \|f_S\|_2^2, \qquad \mathcal{I}_i(f) = 2 \sum_{S \ni i} \|f_S\|_2^2, \qquad \sum_i \mathcal{I}_i(f) \leq 2d \operatorname{Var} f. \tag{5.1}$$

Zero-mass atoms are omitted when forming these L^2 bases.

Proof of Corollary 1.3. At every nonconstant node query a remaining coordinate of largest resampling influence for the conditional function and product law. Constant nodes become leaves. Queries preserve the degree bound. Let ρ_t be the resulting random restriction after depth t, and write

$$D_t = \mathbb{E} \operatorname{Var}(f \mid \rho_t), \qquad P_t = \mathbb{E} \sum_{j \text{ unqueried}} \mathcal{I}_j(f \mid \rho_t).$$

Once a branch ends we pad it with no operation. By (5.1), $P_t \leq 2dD_t$, and D_t is nonincreasing. If the next query at a node is i, then for every $j \neq i$ the average conditional value of $\mathcal{I}_j$ equals its value before that query. This follows by writing its two-copy expectation and first averaging over X_i. Thus

$$P_t - P_{t+1} = \mathbb{E}\mathcal{I}_{i(\rho_t)}(f \mid \rho_t). \tag{5.2}$$

Suppose a stage begins with $D_t \leq u$ and continues while $D_t > u/2$. Let $V = \operatorname{Var}(f \mid \rho_t)$ and $V_+ = V\mathbf{1}_{\{V \geq u/4\}}$. The contribution from $V < u/4$ is at most $u/4$, so $\mathbb{E}V_+ > u/4$. Theorem 1.2 and Jensen's inequality imply

$$P_t - P_{t+1} \geq \frac{c_*}{d^6 b^4} \mathbb{E} V^{5/2} \geq \frac{c_* u^{5/2}}{32 d^6 b^4}.$$

The potential at the beginning of the stage is at most $2du$. Therefore at most $\lfloor 64 d^7 b^4/(c_* u^{3/2}) \rfloor + 1$ levels are needed to make $D_t \leq u/2$.

Start with $u_0 = 1/4$ and halve u until $u \leq \varepsilon^2$. If J stages are needed, then $u_{J-1} > \varepsilon^2$ and

$$\sum_{j=0}^{J-1} u_j^{-3/2} < \frac{\varepsilon^{-3}}{1 - 2^{-3/2}} < 2\varepsilon^{-3}.$$

Since $64 d^7 b^4/(c_* u^{3/2}) > 1$, each extra rounding term can be absorbed by doubling this stage bound. The stated depth follows. At a leaf output $\mathbb{E}[f \mid \rho]$. Its total mean-square error is $\mathbb{E} \operatorname{Var}(f \mid \rho) \leq \varepsilon^2$. If the depth bound exceeds n, querying all inputs gives exact evaluation instead. □

Corollary 5.1. *Let a quantum query algorithm make $T \geq 1$ queries to a finite-alphabet input, with each query returning one symbol, and let $p(x)$ be its acceptance probability on every input. Under every product input law, there exists a deterministic classical tree of depth $O(T^7 b^4 \varepsilon^{-3})$ with mean-square error at most ε^2 in estimating p. For $0 < \eta, \delta < 1$, a depth $O(T^7 b^4 \eta^{-3} \delta^{-3/2})$ suffices for error at most η outside a set of probability at most δ.*

Proof. The polynomial method [3] gives interaction degree at most $2T$. For completeness, each oracle matrix entry is a function of one input symbol. After T queries every amplitude is a sum of terms each depending on at most T input coordinates. Squaring amplitudes and summing over accepting states gives interaction degree at most $2T$. Intermediate measurements can be deferred using an auxiliary register. Corollary 1.3 with $d = 2T$ proves the first statement. For the second

use $\varepsilon = \eta\sqrt{\delta}$ and apply Markov's inequality to the squared estimation error. Values of $\varepsilon \geq 1/2$ can be replaced by a constant-output estimate. □

The tree can use exact conditional expectations and an exhaustive description of the input function. The conclusion controls the number of input queries. It gives no bound on the computation required to construct the tree or its numerical leaf values.

6. The sharp variance question

Proposition 6.1 (The optimal linear case)**.** *Let $f\colon \Omega \to [0,1]$ have interaction degree at most one and variance $v > 0$. Then*

$$\max_i \mathcal{I}_i(f) \geq 8v^2.$$

The constant eight is optimal, even for fair Boolean inputs.

Proof. Write $f = c + \sum_i g_i(x_i)$ and put $r_i = \max g_i - \min g_i$. Independence gives $v = \sum_i v_i$, where $v_i = \operatorname{Var}(g_i(X_i))$, and $\mathcal{I}_i(f) = 2v_i$. Because all combinations of coordinate values belong to the domain, $\sum_i r_i = \max f - \min f \leq 1$. The variance bound for a variable in an interval of length r_i gives $\sqrt{v_i} \leq r_i/2$. If $M = \max_i \mathcal{I}_i(f)$, then

$$v = \sum_i v_i \leq \sqrt{M/2} \sum_i \sqrt{v_i} \leq \frac{1}{2}\sqrt{M/2}.$$

Squaring yields the claim. For $f(x) = n^{-1}\sum_i x_i$ on the fair Boolean cube, $v = 1/(4n)$ and $\mathcal{I}_i = 1/(2n^2) = 8v^2$. □

This example also precludes any influence bound with variance exponent smaller than two and a degree-dependent positive constant. A bound with exponent two and polynomial dependence on arbitrary degree remains an open question; the arguments here give the exponent 5/2.

Appendix A. Analytic estimates and certificate rows

We give the analytic chain from finite interpolation to Proposition 1.5. This also proves the precise pointwise estimate used in Lemma 4.1.

A.1. **Trigonometric interpolation.**

Lemma A.1 (Bernstein's inequality for trigonometric polynomials)**.** *If $H(\theta) = \sum_{j=-D}^{D} h_j e^{ij\theta}$, with $D \geq 1$, then*

$$|H'(\theta)| \leq D \sup_{t\in\mathbb{R}} |H(t)|.$$

Proof. It is enough to evaluate the derivative at zero, since translating H leaves its degree and supremum unchanged. Put

$$t_k = \frac{(2k-1)\pi}{2D}, \qquad c_k = \frac{(-1)^{k-1}}{4D\sin^2(t_k/2)}, \qquad 1 \leq k \leq 2D.$$

We claim that

$$H'(0) = \sum_{k=1}^{2D} c_k H(t_k). \tag{A.1}$$

Here is a finite verification. Orthogonality on this shifted grid shows that the coefficient which differentiates the modes $e^{ij\theta}$ for $-D+1 \leq j \leq D$ is

$$c_k = \frac{1}{2D} \sum_{j=-D+1}^{D} ij e^{-ijt_k}.$$

Indeed, the sum over k of $e^{i(j-r)t_k}$ is zero unless $j = r$ for indices in that interval. The remaining mode $j = -D$ has sampled values equal to the negative of the sampled values of $j = D$, and its derivative at zero is also the negative, so it obeys the same formula.

To compute the coefficient, set $z = e^{-it_k}$, so $z^{2D} = -1$. Differentiating the finite geometric sum gives

$$\sum_{j=-D+1}^{D} jz^j = z\frac{d}{dz}\left(z^{-D+1}\frac{1-z^{2D}}{1-z}\right) = \frac{2z^{-D+1}}{(1-z)^2} \quad \text{at } z^{2D} = -1.$$

Since $(1-z)^2 = -4z\sin^2(t_k/2)$ and $e^{iDt_k} = i(-1)^{k-1}$, this is precisely the displayed value of c_k. This proves (A.1) without an interpolation remainder.

Apply the identity to $H(\theta) = \sin(D\theta)$. Because $\sin(Dt_k) = (-1)^{k-1}$, it gives $\sum_k |c_k| = D$. Taking absolute values in (A.1) now proves the lemma. □

A.2. **A derivative estimate on the solid cube.**

Lemma A.2. *Let $P\colon \mathbb{R}^N \to \mathbb{R}$ be a polynomial of total degree at most d and assume $|P| \le 1$ on $[-1,1]^N$. If $x, y \in [-1,1]^N$, then*

$$|DP(x)[y]| \le d^2. \tag{A.2}$$

Proof. The case $d = 0$ is immediate. For $-1 < r < 1$ consider

$$H_r(\theta) = P\big(rx\cos\theta + \sqrt{1-r^2}\, y\sin\theta\big).$$

Its argument lies in the solid cube. For each coordinate this follows from Cauchy–Schwarz applied to $(r, \sqrt{1-r^2})$ and $(x_i\cos\theta, y_i\sin\theta)$. It is a trigonometric polynomial of degree at most d, with supremum at most one. Thus Lemma A.1 gives

$$\sqrt{1-r^2}\,|DP(rx)[y]| = |H_r'(0)| \le d.$$

The function $g(r) = DP(rx)[y]$ is a polynomial of degree at most $d-1$. Consequently $G(\theta) = \sin\theta\, g(\cos\theta)$ is a trigonometric polynomial of degree at most d and satisfies $|G| \le d$. Apply Lemma A.1 once more. Since $G'(0) = g(1)$, it gives $|DP(x)[y]| \le d^2$. □

The intermediate estimate also explains why the boundary of the cube requires care. The first use of Bernstein loses $(1-r^2)^{-1/2}$ as r approaches one. Multiplication by $\sin\theta$ and the second finite interpolation bound remove that loss. [2]

A.3. **Chebyshev polynomials and overlapping blocks.** Define $T_0(t) = 1$, $T_1(t) = t$, and $T_{m+1}(t) = 2tT_m(t) - T_{m-1}(t)$. The cosine addition identity proves inductively that $T_m(\cos\theta) = \cos(m\theta)$. In particular $\deg T_m = m$, $|T_m| \le 1$ on $[-1,1]$, $T_m(1) = 1$ and $T_m(\cos(\pi/m)) = -1$ for $m \ge 1$.

For a function F on the sign cube with values in $[0,1]$, its multilinear extension $\widetilde{F}$ has the exact formula

$$\widetilde{F}(u) = \sum_{z\in\{-1,1\}^N} F(z)\prod_i \frac{1+z_iu_i}{2}. \tag{A.3}$$

The coefficients in this sum are nonnegative and add to one on the solid cube. Thus its extension still takes values in $[0,1]$ there.

[2]The proof uses total polynomial degree before restricting to vertices. Replacing a Boolean power by its multilinear representative can only decrease that degree.

Lemma A.3 (Block bound)**.** *Let $F\colon \{-1,1\}^N \to [0,1]$ have degree at most D, and fix a vertex x. Let $S_1,\dots,S_k$ be a list of coordinate sets in which each coordinate occurs at most N_0 times. If $m \geq 1$ and $N_0(1-\cos(\pi/m)) \leq 2$, then*

$$\sum_{a=1}^{k} |F(x) - F(x^{S_a})| \leq D^2 m^2. \tag{A.4}$$

Proof. For $w \in \{-1,1\}^k$ put

$$\nu_i(w) = \sum_{a : i \in S_a} \frac{1-w_a}{2}, \qquad \theta_i(w) = T_m(1-(1-\cos(\pi/m))\nu_i(w)),$$

and define $G(w) = \widetilde{F}(x_1\theta_1(w),\dots,x_N\theta_N(w))$. Because $0 \leq \nu_i \leq N_0$, the arguments of every T_m lie in $[-1,1]$. Formula (A.3) implies $0 \leq G \leq 1$. Each θ_i has polynomial degree at most m, so G has degree at most Dm after multilinearization.

At the all-one vertex, every $\theta_i = 1$ and $G(\mathbf{1}) = F(x)$. Changing just coordinate a of w changes θ_i to -1 exactly for $i \in S_a$. Thus $G(\mathbf{1}^{\oplus a}) = F(x^{S_a})$. The pointwise bound of Lemma 4.1, which follows from Lemma A.2, gives (A.4). □

Lemma A.4 (Fractional block packing)**.** *Fix F and x as above. If $\lambda_S \geq 0$ and $\sum_{S \ni i} \lambda_S \leq 1$ for every coordinate i, then*

$$\sum_S \lambda_S |F(x) - F(x^S)| \leq \frac{\pi^2}{4} D^2.$$

Proof. For an integer $M \geq 1$, put $N_0 = M^2$ and $m = \lceil \pi M/2 \rceil$. Make a list in which each block S occurs $\lfloor M^2 \lambda_S \rfloor$ times. Each coordinate occurs at most N_0 times, and

$$N_0(1-\cos(\pi/m)) \leq \frac{M^2\pi^2}{2m^2} \leq 2.$$

Apply Lemma A.3 and divide by M^2. For this fixed finite family the left side converges to the asserted packing sum, while $m^2/M^2 \to \pi^2/4$. □

A.4. **Finite separation and the row estimate.** We include the separation step to make explicit why one row can cover every target input simultaneously.

Lemma A.5. *Let $C \subset \mathbb{R}^m$ be nonempty, compact and convex, with no point in $[0,\infty)^m$. Then there is a nonzero vector $\kappa \geq 0$ such that $\langle \kappa, c \rangle < 0$ for every $c \in C$.*

Proof. Minimize $\rho(c) = \sum_s \max(-c_s, 0)^2$ on C, at a point y. Its minimum is positive. Let $\kappa_s = \max(-y_s, 0)$, so $y + \kappa \geq 0$ and $\langle \kappa, y \rangle = -\rho(y)$. For $c \in C$ and $0 < t \leq 1$, convexity and minimality give

$$\rho(y) \leq \rho(y + t(c-y)) \leq \|t(c-y) - \kappa\|_2^2.$$

Subtract $\rho(y)$, divide by t and let $t \downarrow 0$. It follows that $\langle \kappa, c - y \rangle \leq 0$, proving the claim. □

Proof of Proposition 1.5. Use the sign-cube coordinates, which do not change degree, and fix x. Set $t_S = |F(x) - F(x^S)|$ and $K = \pi^2 D^2/4$. Let $P = \{a \geq 0 : \sum_i a_i \leq K\}$ and consider the compact convex set

$$C = \left\{ \left(\sum_{i \in S} a_i - t_S \right)_{S \subseteq [N]} : a \in P \right\}.$$

If C meets the nonnegative orthant, the corresponding a is the required row. Otherwise Lemma A.5 produces $\kappa_S \geq 0$ with

$$\sum_i a_i \sigma_i < \tau \quad (a \in P), \qquad \sigma_i = \sum_{S \ni i} \kappa_S, \quad \tau = \sum_S \kappa_S t_S.$$

Taking $a = 0$ shows $\tau > 0$. Since $t_\varnothing = 0$, some σ_i is positive. Let j maximize σ_i. The weights $\lambda_S = \kappa_S/\sigma_j$ form a fractional packing, so Lemma A.4 gives $\tau \le K\sigma_j$. Taking $a = Ke_j$ gives the strict reverse inequality, a contradiction. □

Appendix B. The determinant proof of the antipodal inequality

We give Turturean's determinant construction in the notation required for the product transfer. Throughout this appendix all matrices have finite index sets and polynomial entries over $\mathbb{R}$.

B.1. **Agreement monomials and lowest terms.** Let $O \subseteq Q_k$ and let b_j count its boundary edges in direction j. Write $\mathrm{ag}(x, z) = \{j : x_j = z_j\}$ and $s^A = \prod_{j\in A} s_j$. The agreement matrix is

$$M[x, z] = s^{\mathrm{ag}(x,z)}, \qquad x, z \in O.$$

A lowest term of a nonzero polynomial means a monomial with nonzero coefficient of minimum total degree. If a polynomial u has constant term one, multiplication by u leaves the whole lowest homogeneous part of any nonzero polynomial unchanged. We will use this observation when comparing determinants.

Lemma B.1. *The determinant of M is nonzero. Every lowest term s^γ of $\det M$ satisfies $\gamma_j \le b_j$ for all j.*

Proof. If O is empty, its determinant is one. Otherwise specialize every s_j to $c = 2^k$. The diagonal entry of each row is c^k. Each off-diagonal entry is at most c^{k-1}, and there are $|O| - 1 < c$ of them. The resulting real matrix is strictly diagonally dominant, and hence nonsingular. For a direct verification, a nonzero null vector would have a component of maximum absolute value whose row equation contradicts strict dominance. Therefore $\det M$ is nonzero.

Fix j, and let e_j be the number of pairs $\{x, x^{\oplus j}\}$ entirely contained in O. Subtract s_j times the row at $x^{\oplus j}$ from the row at x whenever $x_j = 1$ and both rows belong to O. These operations affect disjoint pairs and leave the determinant unchanged. In each changed row the entries at columns with $z_j = x_j$ vanish. The other entries are $(1 - s_j^2)M[x, z]$. Consequently

$$\det M = (1 - s_j^2)^{e_j} R_j$$

for a polynomial $R_j \neq 0$. Each matrix entry has degree at most one in s_j, so $\deg_{s_j} \det M \le |O| = 2e_j + b_j$. The leading coefficient of a product of nonzero polynomials in s_j cannot vanish. Thus $\deg_{s_j} R_j \le b_j$. Since $(1 - s_j^2)^{e_j}$ has constant term one, the lowest homogeneous parts of $\det M$ and R_j coincide. The asserted exponent bound follows for this j and therefore for every coordinate. □

B.2. **A permutation respecting local subcubes.** Call $A_x \subseteq [k]$ a wing at $x \in O$ when $x^S \in O$ for every $S \subseteq A_x$.

Lemma B.2 (Turturean's permutation construction)**.** *For any wing A_x at every $x \in O$, there is a permutation $\tau\colon O \to O$ such that*

$$\tau(x)_j \neq x_j \quad (j \in A_x), \qquad \#\{x \in O : \tau(x)_j = x_j\} \le b_j \quad (j \in [k]).$$

Proof. Define matrices indexed by O by

$$P[x, z] = \begin{cases} (-1)^{|\Delta(x,z)|} s^{\Delta(x,z)}, & \Delta(x, z) \subseteq A_x, \\ 0, & \text{otherwise}, \end{cases}$$

$$U[x, x] = \prod_{i\in A_x} (1 - s_i^2), \qquad U[x, z] = 0 \quad (x \neq z),$$

$$Q[x, z] = \begin{cases} s^{\mathrm{ag}(x,z)}, & \mathrm{ag}(x, z) \cap A_x = \varnothing, \\ 0, & \text{otherwise}. \end{cases}$$

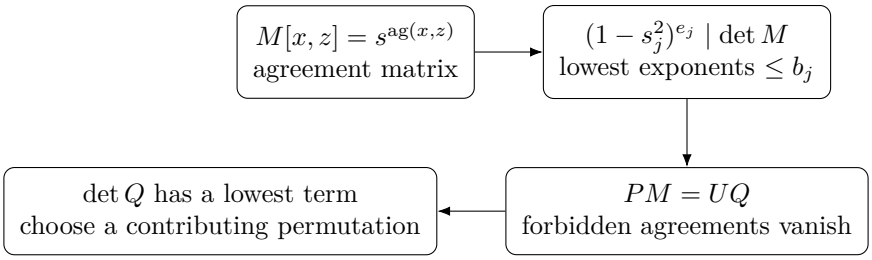


FIGURE 2. The determinant construction supplies one permutation satisfying all coordinate budgets simultaneously. The construction and its antipodal consequence are due to Turturean.

Summing the row combinations of M coordinate by coordinate gives

$$PM = UQ. \tag{B.1}$$

To check an entry, sum over $S \subseteq A_x$ the quantities $(-1)^{|S|} s^S s^{\mathrm{ag}(x^S,z)}$. For $i \in A_x$, its two local terms are $s_i - s_i = 0$ when $x_i = z_i$, and $1 - s_i^2$ otherwise. The remaining coordinates retain their agreement monomials. This is exactly the entry of UQ. The wing property ensures every row x^S used in this sum belongs to O.

At $s = 0$, P is the identity, and each diagonal entry of U is one. Their determinants therefore have constant term one. Taking determinants in (B.1) shows that $\det Q \neq 0$ and that its lowest homogeneous part equals that of $\det M$. Choose a lowest monomial of $\det Q$. The determinant expansion has at least one permutation whose nonzero product of entries contributes to that monomial. Let this permutation be τ. Its nonzero entries ensure $\mathrm{ag}(x, \tau(x)) \cap A_x = \varnothing$. The exponent of s_j in its product is exactly the number of inputs at which $\tau(x)_j = x_j$. Lemma B.1 bounds it by b_j. □

B.3. **From the permutation to rows.** Let $h = \mathbf{1}_O$ and $Z = \{u \in O : \bar{u} \notin O\}$. For positive lengths ℓ_j and a wing A at $u \in Z$, put

$$D(u, A) = \min \left\{ \sum_{j \in G} \ell_j : G \subseteq [k] \setminus A,\ \bar{u}^G \in O \right\}.$$

The minimum exists because $G = [k] \setminus A$ gives $\bar{u}^G = u^A \in O$.

Lemma B.3. *For any wings A_u at $u \in Z$,*

$$\sum_{u \in Z} D(u, A_u) \le \sum_j \ell_j b_j.$$

Proof. Take empty wings on $O \setminus Z$ and apply Lemma B.2. For $u \in Z$, the set $G_u = \mathrm{ag}(u, \tau(u))$ is disjoint from A_u and satisfies $\bar{u}^{G_u} = \tau(u) \in O$. Thus $D(u, A_u) \le \sum_{j \in G_u} \ell_j$. Sum over u, enlarge the sum to O, and use the coordinate budgets of that permutation. □

We also need positivity of the overlap of two Boolean rows. Let $h(u) = 1$, $h(v) = 0$, and let a, b be rows at u, v. Put $D = \Delta(u, v)$ and $S = \{j \in D : a_j \le b_j\}$. The input $u^S = v^{D \setminus S}$ has one of the two Boolean values. If its value is zero, $a(S) \ge 1$. Otherwise $b(D \setminus S) \ge 1$. In both cases

$$\sum_{j \in D} \min(a_j, b_j) \ge 1. \tag{B.2}$$

In particular there is a coordinate with $a_j b_j > 0$.

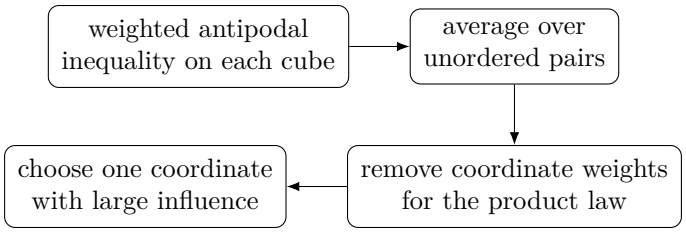


FIGURE 3. The coordinate is selected after the product average. This order prevents a dimension-dependent loss from the changing maximizing index.

Proof of Proposition 1.4. For $u \in Z$, write $a = a^u$, $b = a^{\bar{u}}$ and $\ell_j = w_j^{-1}$. Equation (B.2) ensures $s_u = \sum_j a_j b_j/\ell_j > 0$. Set $\omega_u = s_u^{-1}$ and choose

$$A_u = \{j : \ell_j < \omega_u b_j\}.$$

Because $\sum_j a_j \omega_u b_j/\ell_j = 1$, we have $\sum_{j\in A_u} a_j < 1$. If all those a_j vanish the claim is immediate, and otherwise it follows from the strict inequalities in the definition of A_u. Every subset of A_u therefore leaves the value at u equal to one, so A_u is a wing.

If $G \subseteq [k] \setminus A_u$ and $\bar{u}^G \in O$, the row at $\bar{u}$ gives $\sum_{j\in G} b_j \geq 1$. For each such coordinate $\ell_j \geq \omega_u b_j$, whence $\sum_{j\in G} \ell_j \geq \omega_u$. Thus $D(u, A_u) \geq \omega_u$. Lemma B.3 now gives

$$\sum_{u\in Z} \frac{1}{\sum_j w_j a_j^u a_j^{\bar{u}}} \leq \sum_{u\in Z} D(u, A_u) \leq \sum_j \frac{b_j(h)}{w_j},$$

which is the required inequality. □

APPENDIX C. EXAMPLES AND THE LIMITS OF AVERAGING

C.1. **A rare-coordinate calculation.** For $f(x) = x_1$ under any Bernoulli product law,

$$v = p_1(1-p_1), \qquad \mathcal{I}_1 = 2p_1(1-p_1) = 2v, \qquad \mathcal{I}_i = 0 \quad (i > 1).$$

As $p_1 \to 0$ the variance and the relevant influence vanish together. Theorem 1.2 continues to apply with its fixed constant. At $p_1 = 0$ the variance is zero and no positive lower bound is asserted. [3]

C.2. **A two-coordinate interaction.** For $f(x) = x_1x_2$ with independent probabilities p, q,

$$v = pq(1-pq), \qquad \mathcal{I}_1 = 2p(1-p)q, \qquad \mathcal{I}_2 = 2q(1-q)p.$$

This checks that resampling retains the unchanged coordinate in both copies of the increment. Using a completely independent second input instead would give a different quantity.

C.3. **Why the order of the proof matters.** Applying an influence theorem separately in each conditional cube only gives a lower bound on the average of a maximum. In general this does not lower bound the maximum of the averages. For example, if J is uniform on $[n]$ and $a_i(J) = \mathbf{1}_{\{i=J\}}$, then $\mathbb{E}\max_i a_i(J) = 1$ but $\max_i \mathbb{E}a_i(J) = 1/n$. The argument of Lemma 2.1 avoids this loss by keeping every positive coordinate weight until after averaging. The finite-measure calculation then chooses the weights for the original product law.

[3] A bound uniform in the probabilities need not give a fixed positive influence when its own variance parameter tends to zero. The quantifiers require the displayed variance-dependent estimate at every choice of probabilities.

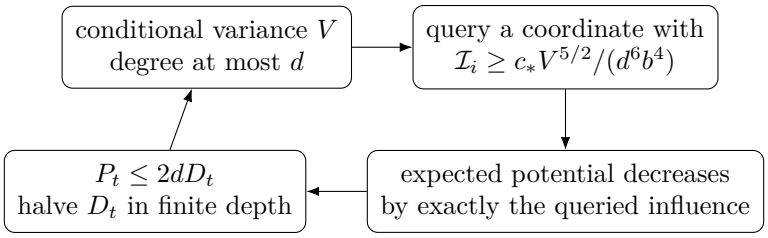


FIGURE 4. The potential is the sum of remaining resampling influences. The identity for its mean decrement supplies a bound on tree depth.

APPENDIX D. VARIANCE CALCULATIONS USED IN THE PROOF

D.1. **Resampling and conditional averaging.** Write $Z = X_{-i}$ and $U = X_i$, and take an independent $V \sim \mu_i$. For a fixed $Z = z$, expansion of the square gives

$$\mathbb{E}_{U,V}(f(z,U) - f(z,V))^2 = 2\mathbb{E}_U f(z,U)^2 - 2(\mathbb{E}_U f(z,U))^2.$$

Thus $\mathcal{I}_i(f) = 2\mathbb{E}_Z \operatorname{Var}_U(f(Z,U))$. If $h(Z) = \mathbb{E}_U f(Z,U)$, expansion around the common mean gives

$$\operatorname{Var} f = \operatorname{Var} h + \mathbb{E}_Z \operatorname{Var}_U(f(Z,U)) = \operatorname{Var} h + \frac{1}{2}\mathcal{I}_i(f).$$

For $j \neq i$, compute the increment of h by using the same averaged U at its two endpoints. The absolute value inequality and Cauchy–Schwarz give, respectively, $\mathcal{L}_j(h) \leq \mathcal{L}_j(f)$ and $\mathcal{I}_j(h) \leq \mathcal{I}_j(f)$. Iterating this argument over a set R yields the exact variance comparison used in Proposition 3.2.

D.2. **The interval variance bound.** For any real random variable $Z \in [a,b]$ with mean m, $(Z-a)(b-Z) \geq 0$. Taking expectations yields $\mathbb{E}Z^2 \leq (a+b)m - ab$, and therefore

$$\operatorname{Var} Z \leq (b-m)(m-a) \leq \frac{(b-a)^2}{4}.$$

This proves both the root bound $\operatorname{Var} f \leq 1/4$ used by the decision tree and the interval estimate in Proposition 6.1.

D.3. **Second and third moments about a median.** For a real random variable Z with finite support, Cauchy–Schwarz twice gives

$$(\mathbb{E}Z^2)^2 \leq \mathbb{E}|Z|\,\mathbb{E}|Z|^3, \qquad (\mathbb{E}|Z|)^2 \leq \mathbb{E}Z^2.$$

After squaring the first inequality and using the second, division by $\mathbb{E}Z^2$ when it is positive gives $(\mathbb{E}Z^2)^3 \leq (\mathbb{E}|Z|^3)^2$. The zero case is immediate. For a median m, one also has $\operatorname{Var} f \leq \mathbb{E}(f-m)^2$, since $\mathbb{E}(f-m)^2 = \operatorname{Var} f + (\mathbb{E}f - m)^2$. These are all the moment comparisons used in (3.4).

APPENDIX E. NORMALIZATION AND DEPENDENCY CHECKS

For $X_i \sim \text{Bernoulli}(p_i)$ and $\Delta_i f = f(x_{-i},1) - f(x_{-i},0)$, direct conditioning gives

$$\mathcal{I}_i(f) = 2p_i(1-p_i)\mathbb{E}_{X_{-i}}(\Delta_i f)^2, \qquad \mathcal{L}_i(f) = 2p_i(1-p_i)\mathbb{E}_{X_{-i}}|\Delta_i f|.$$

These formulas remain valid at $p_i = 0, 1$. At $p_i = 1/2$ the factor is $1/2$. In the $\{-1,1\}$ convention using $\mathbb{E}((f(x) - f(x^{\oplus i}))/2)^2$, the resampling influence is twice that convention's influence.

Appendix B gives the full determinant derivation of Proposition 1.4. Appendix A proves the previously published row bound of Proposition 1.5. All uses of product independence, including conditional orientations, the variance identity and the query potential, are written out above. The supplementary Python file checks the exact rational identities and finite instances; these checks supplement the proofs.

The theorem proved here is the finite-product influence statement with variance exponent $5/2$, and its proof uses no unproved conjecture.

Declarations

Data availability. No data sets were used. The Python program that checks the exact identities and finite instances, together with its output, is included with this preprint as supplementary material and archived with the source of this paper at doi:10.5281/zenodo.24124921.

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Use of artificial intelligence. AI coding assistants were used to write and test the checking program and to check the bibliography. All mathematical arguments were written and verified by the author, who takes full responsibility for the content.

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